

NO.:	G3214-1500	VERSION:	I	PAGE:	1
------	------------	----------	---	-------	---

晶圓廠 阻值量測儀操作規範

Fab resistance measuring instrument practice

壹、目的

I、Purpose

貳、適用時機及適用範圍

II、the timing and scope of application

參、內容

III、contents

一、設備機台(Machine Equipment)

二、操作步驟(Operation Procedure)

肆、操作注意事項

IV、Operation Notes

伍、例行作業

V、Routine Operation

陸、品質控制

VI、Quality Control

柒、儀器 MSA 檢定

VII. The instrument MSA verification

捌、生效與修訂

VIII. Effective versus Revised

NO.:	G3214-1500	VERSION:	I	PAGE:	2
------	------------	----------	---	-------	---

阻值量測儀操作規範 RS-METER MEASUREMENT O.I.

壹、目的 (I. Purpose)

製程目的：利用四點探針法量測控片之阻值以瞭解製程穩定性。

Purpose for Process : We use the 4 pin probe to measure Resistivity (Rs)
of monitor wafer for understanding the stability of process.

貳、適用時機及適用範圍 (II. the timing and scope of application)

N+ 預置，離子植入後之阻值量測。

We use RS-METER to measure Resistivity (RS) of MONITOR WAFER in N+ (POCL3) dopant
process or Implant process.

參、內容 (III. Contents)

一、設備機台

A、Machine Equipment

本機台由 PROMETRIX 公司製造，型號—OMNIMAP RS55。

The machine model is OMNIMAP RS55 produced by PROMETRIX.

二、操作步驟：

B、OPERATION PROCEDURE：

1.開機程序：

POWER ON PROCEDURE

1.1 打開 RS55 機台電源（開關在機台背面左下方）。

Turn on the main power of RS55 (The switch is in back of the machine)

1.2 打開電腦及螢幕之電源。

Turn on the power of computer & monitor screen.

1.3 等待約三分鐘，螢幕上出現如下之畫面。

Wait for about 3 minutes and the computer screen will display as follow：

NO.:	G3214-1500	VERSION:	I	PAGE:	3
------	------------	----------	---	-------	---

Omni Map RS55
with
Stat Trax

COPYRIGHT 1985,86,87,88,89,90 Prometrix CORP.

VERSION:ST-5.20H

FOLDER
SELECT

LOG ON
NEW DISK

TURN
DEMO ON

PUT
STAGE DN

PUT
STAGE UP

SET UP

2.按 F1(FOLDER SELECT)，即進入以下螢幕，此時置放晶片之平台升起，以真空吸筆將晶置於平台上，大平邊朝自己，晶片外圍與平台上兩個鐵氟龍螺絲相接。

Press F1(FOLDER SELECT)；Then the screen will show as follow; The stage will go up at this time. Put the wafer on stage with sucking pen.The flat of wafer face to yourself. The edge of wafer touch Teflon screw on stage.

NO.:	G3214-1500	VERSION:	I	PAGE:	4
------	------------	----------	---	-------	---

FOLDER ID:115-Q 597

CABINET	DRAWER	FOLDER
DIFFUSION IMPLAATER	D65 D54 D30 RTP 925C	10.5± 1 ohm/s 17± 1 ohm/s 37± 2 ohm/s 42± 3 ohm/s 45± 4 ohm/s

▲

▼

▲

▼

▲

▼

COLLECT
NEW DATA

COMBINE
OLD DATA

CHANGE
PROBE

TREND
CHART

SQC
CHART

CURVE
FITTING

FOLDER
DRCTRY

INTRO

3.以左右方向鍵將反白區移至檔名欄 (CABINET)，再以上下方向鍵選擇檔名，以同樣方式依序選擇機台 (DRAWER) 及規格 (FOLDER)。

To choose CABINET / DRAWER / FOLDER filename with direction key (UP/DOWN/LEFT/RIGHT).

4.確定以上所選項目後，按 F1(COLLECT NEW DATA) 進入以下螢幕，再按 F1(LOAD WAFER) 即開始量測。

To make sure the item you choose is right. Press F1 (COLLECT NEW DATA) to next screen. Now press F1 (LOAD WAFER) and it will begin to measure resistivity of this monitor wafer.

NO.:	G3214-1500	VERSION:	I
		PAGE:	5

Test Type

Current Value

Card Index

Wafer Setup

Test Setup

PgUp
PgDn

TITEL..

TEST TYPE.....	QUICK TEST
TEST SITES.....	5
WAFER DIAM.....	150.00mm/5.91in
TEST DIAM.....	130.00mm/5.12in
SORTING SIGMA.....	3.0
QUICK TEST.....	PROMETRIX 5
ROTATION.....	0 degrees
AUTO SAVE.....	AUTO SAVE,NO UPLADE
CORRELATION.....	NONE
UNITS.....	OHMS/SQ

LOAD WFR

EDIT ACTIVE

TOGGLE ACTIVE

NEW TEST

FOLDER SELECT

5.量測完畢則出現如以下螢幕之數據摘要 (DATA SUMMARY)，記錄所要之數據後，此時機台會自動存入數據，如須以同樣檔案量測下一片晶片，則以同樣方式放晶片於平台上後，按 F6 (NEW TEST) 繼續量測，如不量則按 F8 三次回到主螢幕，再按 F5 (PUT STAGE DOWN) 後，平台降下回復準備位置。

The Computer will display DATA SUMMARY in monitor screen when measurement is complete. Then you can record Resistivity DATA .And computer will AUTO-SAVE DATA when measure complete. If you want to measure next wafer with the same measurement file, you may press F6 (NEW TEST) key to continue measuring after you put next monitor wafer on stage. If you do not want to measure or to measure with other file,then press F8 key three times to quit to main screen for selecting new file. In main screen you must press F5 (PUT STAGE DOWN) key

NO.:	G3214-1500	VERSION:	I	PAGE:	6
------	------------	----------	---	-------	---

to put the stage down when you do not want to measure any wafer.

Card Index Wafer Setup Test Setup Test Results Trend Charts SQC Charts PgUp PgDn	File Summary Data Summary TITEL.. <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="border: 1px solid black; padding: 2px;">MEAN.....</td> <td style="text-align: right; padding: 2px;">44.786</td> </tr> <tr> <td>STD DEV.....</td> <td style="text-align: right;">1.360 3.035%</td> </tr> <tr> <td>MINIMVN.....</td> <td style="text-align: right;">43.440</td> </tr> <tr> <td>MAXIMVN.....</td> <td style="text-align: right;">47.050</td> </tr> <tr> <td>RANGE.....</td> <td style="text-align: right;">3.610</td> </tr> <tr> <td># SITES/GOOD.....</td> <td style="text-align: right;">5/ 5</td> </tr> <tr> <td>SORTING SIGMA.....</td> <td style="text-align: right;">3.0</td> </tr> <tr> <td>CORRELATION.....</td> <td style="text-align: right;">*NONE</td> </tr> <tr> <td>UNITS.....</td> <td style="text-align: right;">OHMS/SQ</td> </tr> </table> ▲ ▼	MEAN.....	44.786	STD DEV.....	1.360 3.035%	MINIMVN.....	43.440	MAXIMVN.....	47.050	RANGE.....	3.610	# SITES/GOOD.....	5/ 5	SORTING SIGMA.....	3.0	CORRELATION.....	*NONE	UNITS.....	OHMS/SQ
MEAN.....	44.786																		
STD DEV.....	1.360 3.035%																		
MINIMVN.....	43.440																		
MAXIMVN.....	47.050																		
RANGE.....	3.610																		
# SITES/GOOD.....	5/ 5																		
SORTING SIGMA.....	3.0																		
CORRELATION.....	*NONE																		
UNITS.....	OHMS/SQ																		

DATA
VALUES

NEW TEST

FOLDER
SELECT

肆、操作注意事項：(IV. OPERATION NOTES：)

一、檔案 (CABINET) 使用說明：

A、EXPLANATION FOR CABINET OPERATION：

1.DIFFUSION：N+ 預置爐管及RTP機台 Rs 測量專用。

DIFFUSION：The file which could measure Resistivity of monitor wafer for
N+ dopant Furnace & RTP machine.

2.IMPLANTER：IMPLANTER Rs 測量專用。

IMPLANTER：The file which can measure Resistivity of monitor wafer for
IMPLANTER.

NO.:	G3214-1500	VERSION:	I	PAGE:	7
------	------------	----------	---	-------	---

二、螢幕畫面左上角出現 DEMO MODE 紅色字樣時，按 F4(TURN DEMO ON) 鍵直到 DEMO MODE 字樣消失，才可以量測。

B、When the computer screen show "DEMO MODE" on the up-left corner,you can press F4 (TURN DEMO ON) key and waitting for the "DEMO MODE" disappear.Then you can measure monitor wafer until the "DEMO MODE" disappear.

三、Rs-55 機台之液晶顯示幕出現 VACUUM AIR IS INSUFFICIENT 時，停止測量並通知設備檢查真空壓力是否足夠。

C、If RS-55 LCD screen show "VACUUM AIR IS INSUFFICIENT",you must stop measurement. And contact with EQ to check vacuum pressure.

四、畫面不動，而 Rs-55 機台門無法自動開啟時：

D、When the screen is locked and the door of RS-55 can't open：

1.按 Rs-55 之 RESET 鍵，若液晶顯示幕出現 VER 2.10 09-08-89 時，則可測量。

Press "RESET" key on RS-55. Waitting for the RS-55 LCD show "VER 2.10 09-08-89". Then You can continue measurement.

2.若液晶顯示幕仍無 VER 2.10 09-08-89 字樣，則將 RS-55 之電源關掉，過五秒後再開啟，若可出現 VER 2.10 09-08 89 字樣則可測量。

If the RS-55 LCD still can't show "VER 2.10 09-08-89".Then turn-off power of RS-55 waiting for 5 seconds. And turn-on the power of RS-55 after 5 seconds. Until the RS-55 LCD show "VER 2.10 09-08-89",then you can continue measurement.

3.上述步驟無效時，則通知設備工程師處理。

If it still can't work after you do all operation step above. Please contact with EQ to check hardware.

伍、例行作業 (V. ROUTINE OPERATION)

一、每天日班負責 N+ 預置 (D54) 之 EA，於上班初以 IPA 擦拭機台。

A、EA who charge F25 (D54) in Day-class must to wipe RS-METER with IPA lightly every morning.

陸、品質控制 (VI. QUALITY CONTROL)

一、儀器定期校驗：(I. Instrument periodical calibration：)

1.校驗執行人：爐管 PE。

Calibration Operator：DIFFUSION MODULE PE.

校驗頻率：6 個月。

NO.:	G3214-1500	VERSION:	I	PAGE:	8
------	------------	----------	---	-------	---

Calibration frequency : 6 months

標準件：PROMETRIX 提供之標準電阻。

Standard device : The standard Resistivity which is provided by PROMETRIX.

存放地點：置於晶圓廠內，溫濕度同 FAB 條件。

Storage : located in FAB the environment is the same as FAB.

- 校驗程序：取下一般測值用之 PROBE 之後，插上標準電阻，按照本規範前述之操作步驟，選取 "AUTO TEST" 檔名後開始量測。

Calibration procedure Remove: the normal measurement PROBE. Put the standard resistivity on head. Then follow the operation procedure B-3,4 above. Selecting "AUTO TEST" begin to measure.

機台校驗規格：(Calibration SPEC.)

請上"儀器校驗報告書電子表單系統 http://nthcweb01/CMS/CMS_EFORM/ " 查詢。(範例如下)

Please refer to the "Calibration Report Form"http://nthcweb01/CMS/CMS_EFORM/ Inquire.
(Example below)

項次: 1 校驗項目: Sheet Resistivity (RS)					
儀器使用(點)範圍: 2.79 / 279 / 27.9K Ω/ sq					
允收規格: ± 1 %			規格計算方式: ABS(B2-A)/A*100%		
測量(點)範圍					
item	標準件-讀值(A)	被校件-設定(B1)	被校件-讀值(B2)	單位	備註
1	2.79	NA		Ω/ sq	
2	279	NA		Ω/ sq	
3	27900	NA		Ω/ sq	
校驗時使用之標準件或輔助儀器-第1組					
儀器編號	儀器名稱	製造廠商	型號	序號	
S210-003	阻值量測標準件(精密電阻) (E)	PROMETRIX	STD-RS	S23-SR-3PCS	

- 校驗報告傳至儀校管理單位。

The original of calibration report will be forwarded to Instrument Calibration Center.

- 校驗結果不合規格時，由 S210 人員會同 PROMETRIX 或代理商人員檢修，重新校驗結果合規格後，機台始能繼續上線量測，遇有標準件失效破損則由 PE 申請更換。

When the data out of spec, S210 must contact with PROMETRIX or VENDER to check. And calibrate again. Until the data is in spec , the machine (RS-METER) can continue to measure on line. If the Standard Device fail or broken, a renewal application should be submitted by PE.

- 每 6 個月須作 GRR 並會由 儀校管理單位 存查。

Every six months do GRR and data keep in QC.

二、標準件定期校驗：

B、STANDARD DEVICE periodical calibration：

- 校驗執行人：儀校管理單位。

NO.:	G3214-1500	VERSION:	I	PAGE:	9
------	------------	----------	---	-------	---

Calibration Operator : QC

校驗頻率：一年。

Calibration frequency : ONE YEAR

標準件：PROMETRIX 提供之標準電阻。

Standard device : The standard Resistivity which is provided by PROMETRIX.

2.標準件校驗規格

STANDARD DEVICE CALIBRATION SPEC.

請上”儀器校驗報告書電子表單系統 http://nthcweb01/CMS/CMS_EFORM/ “查詢。(範例如下)

Please refer to the "Calibration Report Form"http://nthcweb01/CMS/CMS_EFORM/"Query (example below)

詳細規格					
項次：1 校驗項目：STD Sheet Resistivity (STD RS)					
儀器使用(點)範圍：2.79 Ω/sq					
允收規格：± 0.3 %					
規格計算方式：ABS(B1-A)/B1*100%					
測量(點)範圍					
item	標準件-讀值(A)	被校件-設定(B1)	被校件-讀值(B2)	單位	備註
1		2.79	NA	Ω/ sq	0.1V
2		2.79	NA	Ω/ sq	0.5V
3		2.79	NA	Ω/ sq	1V
項次：2 校驗項目：STD Sheet Resistivity (STD RS)					
儀器使用(點)範圍：279 Ω/sq					
允收規格：± 0.3 %					
規格計算方式：ABS(B1-A)/B1*100%					
測量(點)範圍					
item	標準件-讀值(A)	被校件-設定(B1)	被校件-讀值(B2)	單位	備註
1		279	NA	Ω/sq	1V
2		279	NA	Ω/sq	5V
3		279	NA	Ω/sq	10V
項次：3 校驗項目：STD Sheet Resistivity (STD RS)					
儀器使用(點)範圍：27.9 KΩ/sq					
允收規格：± 0.3 %					
規格計算方式：ABS(B1-A)/B1*100%					
測量(點)範圍					
item	標準件-讀值(A)	被校件-設定(B1)	被校件-讀值(B2)	單位	備註
1		27.9	NA	KΩ/sq	1V
2		27.9	NA	KΩ/sq	5V
3		27.9	NA	KΩ/sq	10V

3.校驗報告傳至儀校管理單位。

The original of calibration report will be forwarded to Instrument Calibration Center.

4.校驗結果不合規格時，請參照 儀器校驗管理規範處理，處理完重新校驗結果合規格後，機台始能繼續上線量測。

If calibration data is out of SPEC. Then follow INSTRUMENT CALIBRATION O.I of

QC. Until the data is in spec, the machine (RS-METER) can continue to measure on line.

三、定期儀器之校正時機依儀器經歷卡上之校正週期執行校正，量測標準片比對量測值與標準值並記錄於校正報告上，依經歷卡上的容許誤差值進行是否校正合格之判定，並依

NO.:	G3214-1500	VERSION:	I	PAGE:	10
------	------------	----------	---	-------	----

“0000-1000 儀器校驗管理程序” 執行相關校驗規定。

3. The calibrating time of the periodic instrument is calibrated according to the calibration period on the instrument experience card. The measured standard value is compared with the standard value and recorded on the calibration report. According to the permissible error value on the experience card, And in accordance with the "0000-1000 instrument calibration management procedures," the implementation of the relevant validation requirements.

柒、儀器 MSA 檢定 VII. the instrument MSA verification

本儀器納入量具 MSA 檢定系統，依量具 MSA 經歷卡之檢定週期進行檢定，檢定之儀器操作同儀器之操作程序。

The instrument into the measuring instrument MSA verification system, according to the amount of MSA experience card verification cycle for verification, verification of the instrument operation with the instrument operating procedures

一、I. GRR

儀器定期校驗合格後，取合乎量具 MSA 經歷卡上 SAMPLE 規格之 Bare WAFER，從每班工程助理中輪流各擇取一人，以新進人員為優先，依經歷卡上 GRR 各項規定內容進行檢定。

After regular calibration, the Bare WAFER of the SAMPLE specification on the MSA experience card is selected. In turn, each of the engineering assistants will select one of them. The new personnel will be the priority, and the GRR will be verified according to the requirements of the GRR.

二、II. Bias/Linearity

依量具 MSA 經歷卡上所規定之校驗合格標準片，連續進行量測後，進行數據的分析與檢定。

三、MSA 相關作業依規範 “4040-0114 MSA(量測系統分析)作業規範” 執行。

III. MSA-related operations are performed in accordance with the specification "4040-0114 MSA (Measurement System Analysis) Operation Specification".

四、MSA 檢定規格 (MSA Test Spec)：

IV, MSA test specifications (MSA Test Spec):

請上”儀器校驗報告書電子表單系統 http://nthcweb01/CMS/CMS_EFORM/ “查詢。(範例如下)

Please refer to the "Calibration Report Form"http://nthcweb01/CMS/CMS_EFORM/"Query (example below)

NO.:	G3214-1500	VERSION:	I	PAGE:	11
------	------------	----------	---	-------	----

%GRR詳細規格								
檢定項目	樣本數	(晶片) 測試點	重複次數	作業員人數	製程規格			
					最小刻度	規格下限	規格上限	單位
RS	5	5	3	3	0.001Ω/□	9.5	11.5	Ω/□
%BIAS詳細規格								
檢定項目		製程規格						
	標準片值		最小刻度		規格下限		規格上限	
RS	2.79		0.001Ω/□		9.5		11.5	
%Linearity詳細規格								
檢定項目		測量範圍			標準片值		單位	
RS		10.5~45			2.79		Ω/□	
					279			
					27900			

捌、生效與修訂

VIII. Effective versus Revised

本規範之公佈實施及其修訂核准層級皆依會簽/核決/分發依循範例為之。

The implementation of this Code and its revised approval levels are governed by the terms of signature / approval / distribution